MORPHISMS FROM AN AFFINE SCHEME OF A LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2558-2581, following the statement at 2549-2556.

Allowed standard prerequisites (editorial).

- Spec(Z) is final among locally ringed spaces / morphisms into affines. lemma-morphism-into-affine (same chapter, cited at ~2564). Reduces the affine case of X to the ring-level factorization.

Source reference. Original lemma 01I1: sources/stacks-schemes/pinned-original-source.tex, source lines 1058–1058. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Let R be a local ring. The construction above gives a bijective correspondence between morphisms $\mathrm{Spec}(R) \to X$ and pairs (x, φ) consisting of a point $x \in X$ and a local homomorphism of local rings $\varphi : \mathcal{O}_{X,x} \to R$.*

Proof. Let A be a ring. For any ring homomorphism $\psi : A \to R$ there exists a unique prime ideal $\mathfrak{p} \subset A$ and a factorization $A \to A_{\mathfrak{p}} \to R$ where the last map is a local homomorphism of local rings. Namely, $\mathfrak{p} = \psi^{-1}(\mathfrak{m})$. Via Lemma 01I1 this proves that the lemma holds if X is an affine scheme.

Let X be a general scheme. Any $x \in X$ is contained in an open affine $U \subset X$. By the affine case we conclude that every pair (x, φ) occurs as the end product of the construction above the lemma.

To finish the proof it suffices to show that any morphism $f : \mathrm{Spec}(R) \to X$ has image contained in any affine open containing the image x of the closed point of $\mathrm{Spec}(R)$. In fact, let $x \in V \subset X$ be any open neighbourhood containing x. Then $f^{-1}(V) \subset \mathrm{Spec}(R)$ is an open containing the unique closed point and hence equal to $\mathrm{Spec}(R)$. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.